

Chad Carmichael

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EDUCATION

Mechanical Engineering Florida State University

Expected May 2028

Vires Scholar — merit scholarship, \$16,000 over 8 semesters. | **IB Diploma** recipient, Deerfield Beach High School.

EXPERIENCE/ PROJECTS

TurfTrack / FairLie — Smart Golf Training Mat

- Architected a **3-board embedded PCB (ESP32-S3, XIAO nRF52840, Pro Micro nRF52840)** fusing **6 pressure sensors**, a 9-axis IMU, and a 60GHz radar module to capture real-time swing and ball-strike data
- **Debugged** communication, sensor, and **integration issues** across **multiple microcontrollers** and **embedded sensor modules** during prototype bring-up.
- Developed **Python** scripts to **process, visualize, and validate** sensor telemetry during system testing.

Personal NAS Server

Mini-ITX Build · SolidWorks · Cursor

- Designed and assembled a custom **Mini-ITX NAS** system with 8 TB of storage, **integrating motherboard, storage, cooling, power, and enclosure constraints**.
- **Designed the custom** enclosure in **SolidWorks** around component **clearances, airflow, cable routing, and serviceability**.
- **Developed** software to automatically ingest and organize files from removable storage.

Stirling Engine Design & Assembly — SolidWorks

- **Modeled and assembled** a multi-component Stirling engine in SolidWorks from **engineering drawings**.
- **Designed threaded, toleranced, and interfacing components** and produced assembled and exploded technical drawings with BOM documentation.
- **Modeled and assembled a multi-component V8 engine** in SolidWorks from engineering drawings, including **pistons, connecting rods, crankshaft, cylinder block, and related hardware**.
- Created **detailed** part models and constrained assemblies while **accounting for component fit, alignment, and mechanical motion**.
- **Produced engineering drawings** and exploded assembly views to **document** component relationships and assembly sequence.

SKILLS

Skills — Embedded Systems: ESP32-S3, nRF52840, BLE, sensor integration, embedded electronics • **Programming:** C++, Python, MATLAB • **Hardware:** PCB prototyping, PCB fabrication, hardware integration, soldering, debugging • **Interfaces:** I²C, SPI, UART • **CAD:** SolidWorks, Autodesk, KiCAD, Fusion 360, Inventor • **Sensors:** IMU, pressure sensors, radar sensors • **Tools:** Git, Cursor, Microsoft Excel

LEADERSHIP AND BUSINESS VENTURES

Co-Founder — EZclickAI & C2x Visuals

- **Scaled client apps and personal brands** to **1M+ combined online views** through short-form content strategy, editing, and distribution
- **Grew one client's revenue by 50% in a single month** by rebuilding their brand presence and content pipeline
- **Collaborate** with FSU Unconquered Media

SOFTWARE DEVELOPMENT

- **Engineered and deployed** a full-stack iOS application using React Native and Expo, **implementing** a swipe-based UI to modernize university discovery.
- **Assisted** in architecting an **automated** data pipeline using the ChatGPT API (LLMs) to clean and normalize a high-volume dataset of **300,000+** raw records, **transforming** them into a structured repository of **1,000+** validated profiles for analysis.

CLUBS

- National Society of Black Engineers (NSBE) — Member/ NextGen committee
- SMILI (Student Minority in Mentor Leadership Initiative) — Member
- ASCE/FES (American Society of Civil Engineer/ Florida Engineering Society) - 3D Printing Team